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DISPLAY APPARATUS AND HOME APPLIANCE HAVING SAME

Abstract

A home appliance including a display apparatus configured to project an image including state information of the home appliance. The display apparatus includes an image generator including a light source, a display device configured to transmit light generated by the light source and display the image, and an illumination lens configured to condense the light generated by the light source to the display device, an adjustor including a refractive lens configured to adjust a size of the image projected from the display apparatus, and a housing configured to accommodate the image generator and the adjustor, wherein the light source, the illumination lens, the display device, and the refractive lens are sequentially arranged along an optical axis of the image generator, and the refractive lens is configured to move along the optical axis relative to the light source, the illumination lens, and the display device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
[0001] This application is a continuation application is a continuation application, under 35U.S.C. § 111(a), of international application No. PCT/KR2023/019789, filed Dec. 4, 2023, which claims priority under 35 U. S. C. § 119 to Korean Patent Application No. 10-2022-0190440, filed Dec. 30, 2022, and Korean Patent Application No. 10-2023-0075454, filed Jun. 13, 2023, the disclosures of which are incorporated herein by reference in their entireties.
TECHNICAL FIELD
[0002] The disclosure relates to an apparatus configured to display a display by a beam projector and a home appliance including the apparatus.
BACKGROUND ART
[0003] Home appliances are mainly provided in a user's home to help with the user's housework, and include refrigerators, air conditioners, air purifiers, vacuum cleaners, cooking appliances, dishwashers, clothes care machines, washing machines, and the like.
[0004] The home appliance may include a display apparatus configured to allow a user to check a driving state of the home appliance.
[0005] The display apparatus may display a display through a display device arranged in the home appliance or display a display by projecting light on an outer region of the home appliance through a beam projector function (image projection function).
[0006] DISCLOSURE
Technical Problem
[0007] An embodiment of the present disclosure provides a display apparatus capable of projecting an image, the display apparatus capable of allowing an image generator, which is configured to provide light and generate an image, and an adjustor, which is configured to adjust a projected size of the image generated by the image generator, to be a single module, and a home appliance including the same.
Technical Solution
[0008] According to an embodiment of the present disclosure, a home appliance includes a display apparatus configured to project an image including state information of the home appliance. The display apparatus includes an image generator including a light source, a display device configured to transmit light generated by the light source and display the image, and an illumination lens configured to condense the light generated by the light source to the display device, an adjustor including a refractive lens configured to adjust a size of the image projected from the display

apparatus, and a housing configured to accommodate the image generator and the adjustor, wherein the light source, the illumination lens, the display device, and the refractive lens are sequentially arranged along an optical axis of the image generator, and the refractive lens is configured to move along the optical axis relative to the light source, the illumination lens, and the display device.
Advantageous Effects
[0009] The display apparatus according to various embodiments of the present disclosure may minimize a size of the module including the image generator and the adjustor.

Description

DESCRIPTION OF DRAWINGS
[0010] FIG. **1** is a view illustrating a system kitchen in which a dishwasher according to an embodiment is installed in a built-in manner.
[0011] FIG. **2** is a perspective view of a state in which a door of the dishwasher according to an embodiment is opened.
[0012] FIG. **3** is a schematic cross-sectional view of the dishwasher according to an embodiment.
[0013] FIG. **4** is a control block diagram of the dishwasher according to an embodiment.
[0014] FIG. **5** is a conceptual diagram of a display apparatus according to an embodiment.
[0015] FIG. **6** is a perspective view of the display apparatus according to an embodiment.
[0016] FIG. **7** is a perspective view of the display apparatus according to an embodiment.
[0017] FIG. **8** is a cross-sectional view of the display apparatus according to an embodiment.
[0018] FIG. **9** is an exploded perspective view of a housing of the display apparatus according to an embodiment.
[0019] FIG. **10** is an exploded perspective view of a second housing of the display apparatus according to an embodiment.
MODES OF THE INVENTION
[0020] The various embodiments and the terms used therein are not intended to limit the technology disclosed herein to specific forms, and the disclosure should be understood to include various modifications, equivalents, and/or alternatives to the corresponding embodiments.
[0021] In describing the drawings, similar reference numerals may be used to designate similar constituent elements.
[0022] A singular expression may include a plural expression unless they are definitely different in a context.
[0023] The expressions "A or B," "at least one of A or/and B," or "one or more of A or/and B," "A, B or C," "at least one of A, B or/and C," or "one or more of A, B or/and C," and the like used herein may include any and all combinations of one or more of the associated listed items.
[0024] The term of "and/or" includes a plurality of combinations of relevant items or any one item among a plurality of relevant items.
[0025] Herein, the expressions "a first", "a second", "the first", "the second", etc., may simply be used to distinguish an element from other elements, but is not limited to another aspect (e.g., importance or order) of elements.
[0026] When an element (e.g., a first element) is referred to as being "(functionally or communicatively) coupled," or "connected" to another element (e.g., a second element), the first element may be connected to the second element, directly (e.g., wired), wirelessly, or through a third element.
[0027] In this disclosure, the terms "including", "having", and the like are used to specify features, numbers, steps, operations, elements, components, or combinations thereof, but do not preclude the presence or addition of one or more of the features, numbers, steps, operations, elements, components, or combinations thereof.

[0028] When an element is said to be “connected”, “coupled”, “supported” or “contacted” with another element, this includes not only when elements are directly connected, coupled, supported or contacted, but also when elements are indirectly connected, coupled, supported or contacted through a third element.
[0029] Throughout the description, when an element is “on” another element, this includes not only when the element is in contact with the other element, but also when there is another element between the two elements.
[0030] Hereinafter exemplary embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.
[0031] The present disclosure describes a dishwasher **1** as an example of home appliances, but it may be applied to various home appliances such as washing machines, refrigerators, ovens, microwave ovens, and air conditioners.
[0032] On the other hand, the terms “up and down”, “lower side”, and “front and rear” used in the following description are defined based on the direction arrows of FIG. **1**, but the shape and location of each component are not limited by these terms.
[0033] FIG. **1** is a view illustrating a system kitchen in which a dishwasher according to an embodiment is installed in a built-in manner, FIG. **2** is a perspective view of a state in which a door of the dishwasher according to an embodiment is opened, and FIG. **3** is a schematic cross-sectional view of the dishwasher according to an embodiment. Referring to FIGS. **1** to **3**, a system kitchen may include a cabinet S including a storage space R. The cabinet S may include a counter arranged on the cabinet S in a flat plate shape and capable of allowing a user to cook, and a sink provided to wash dishes or prepare ingredients.
[0034] The storage space R may be partitioned by internal partition walls, etc., and the partitioned storage space R may be opened and closed by various panels P forming a front surface of the cabinet S.
[0035] In the system kitchen, a home appliance may be provided in a built-in manner.
[0036] For example, the dishwasher **1** may be provided in the system kitchen in a built-in manner. The dishwasher **1** may be arranged in a part of the partitioned storage space R, and the storage space R in which the dishwasher **1** is arranged may be opened and closed by a front panel **90** arranged in the front surface of the dishwasher **1** among the various panels P.
[0037] For example, other than the dishwasher **1**, other types of home appliances such as an oven and a microwave oven may be stored in the storage space R.
[0038] For example, the dishwasher **1** may be disposed in a storage space formed inside an art wall of an indoor space other than the system kitchen, and the storage space, in which the dishwasher **1** is stored, may be opened and closed by a panel forming the art wall.
[0039] For example, other than the dishwasher **1**, other types of home appliances such as an air conditioner, a washing machine, a clothes dryer, an oven, and a microwave oven may be stored in the storage space formed inside the art wall.
[0040] For example, the dishwasher **1** may be disposed in an inner space formed inside a storage cabinet such as a furniture other than the system kitchen or the art wall, and the storage space in which the dishwasher **1** is arranged may be opened and closed by a door of the storage cabinet.
[0041] For example, other than the dishwasher **1**, other types of home appliances such as an air conditioner, a washing machine, a clothes dryer, an oven, and a microwave oven may be stored in the storage space formed inside the storage cabinet.
[0042] However hereinafter the dishwasher **1** stored in the system kitchen will be described as an example, and the following features may be applied to home appliances stored in various forms described above.
[0043] The dishwasher **1** may include a tub **12** provided inside a main body **10**. The tub **12** may be provided in a substantially box shape. One side of the tub **12** may be open. The tub **12** may include an opening **12***a*. As an example, the tub **12** may open toward a first direction X, which is the front.

[0044] The dishwasher **1** may further include a door **20** configured to open and close the opening **12***a* of the tub **12**. The door **20** may be installed on the main body **10** to open and close the opening **12***a* of the tub **12**. The door **20** may be rotatably installed on the main body **10** through a member such as a hinge **25**. The door **20** may be detachably mounted to the main body **10**.
[0045] For example, the door **20** may be rotatably hinged to a lower portion of the main body **10**. An axis of rotation of the hinge **25** may extend in a second direction Y, which is a left and right direction of the main body **10**, and thus the door **20** may be rotated in a front and rear direction from the front of the main body **10**.
[0046] For example, the door **20** may be hinged to a hinge provided on the left or right side of the main body **10** in the second direction Y on the front side of the main body **10**. The door **20** may be provided to be rotated from the second direction Y to the first direction X by hinges arranged on the left and/or right side of the main body **10**.
[0047] The door **20** may include an outer surface **22** forming the outer side of the door **20** and coupled to the front panel **90**, and an inner surface **21** provided to face the inside of the tub **12** when the door **20** closes the tub **12**.
[0048] The dishwasher **1** may further include a storage container provided inside the tub **12** to store dishes. The storage container may include a plurality of baskets **51**, **52**, and **53**.
[0049] The storage container may include an intermediate basket **52** positioned in the middle with respect to the height direction of the dishwasher **1**, and a lower basket **51** positioned in a lower portion with respect to the height direction of the dishwasher **1**. The intermediate basket **52** may be provided to be supported by an intermediate guide rack **13***b*. The lower basket **51** may be provided to be supported by a lower guide rack **13***a*. The intermediate guide rack **13***b* and the lower guide rack **13***a* may be installed on a side surface **12***d* of the tub **12** so as to be slidable toward the opening **12***a* of the tub **12**. The side surface **12***d* of the tub **12** may include an inner surface of a right wall and an inner surface of a left wall of the tub **12**.
[0050] Relatively large dishes may be stored in the lower basket **51** and the intermediate basket **52**. However, the types of dishes accommodated in the lower and intermediate baskets **51** and **52** is not limited to relatively large dishes. That is, the plurality of baskets **51**, **52** and **23** may accommodate not only relatively large dishes but also relatively small dishes.
[0051] The storage container may include an upper basket **53** positioned in an upper portion with respect to the height direction of the dishwasher **1**. The upper basket **53** may be formed in a rack assembly to accommodate relatively small dishes. For example, the upper basket **53** may accommodate a cooking utensil such as a ladle, a knife, or a turner, or cutlery. In addition, the rack assembly may accommodate a small cup such as an espresso cup. However, the types of dishes accommodated in the upper basket **53** is not limited thereto.
[0052] The upper basket **53** may be provided to be supported by an upper guide rack **13***c*. The upper guide rack **13***c* may be installed on the side surface **12***d* of the tub **12**. For example, the upper basket **53** may be slidably moved by the upper guide rack **13***c*, and inserted into or withdrawn from a washing chamber C.
[0053] The storage container is not limited to the shape shown in FIGS. **2** and **3**, and the storage container may not include the upper basket **53** according to the size of the tub **12**. For example, the storage container may be implemented with the intermediate basket **52** and the lower basket **53**.
[0054] The dishwasher **1** may include the washing chamber C, which is a space formed inside the tub **12**. The washing chamber C may be defined as an inner space of the tub **12**. The washing chamber C may correspond to a space surrounded by a lower surface **12***b*, an upper surface **12***c* and the side surface **12***d* of the tub **12**, and the inner surface **21** of the door **20** when the door **20** closes the tub **12**.
[0055] The washing chamber C may refer to a space in which dishes placed in the baskets **51**, **52** and **53** are washed by wash water and dried.
[0056] The dishwasher **1** may include a spray device **40** configured to spray wash water. The spray

device **40** may receive wash water from a sump assembly **70**.
[0057] The spray device **40** may include a plurality of spray units **41**, **42**, and **43**.
[0058] For example, the plurality of spray units **41**, **42**, and **43** may include a first spray unit **41** arranged under the lower basket **51** in the height direction of the dishwasher **1**, a second spray unit **42** arranged under the intermediate basket **52** in the height direction of the dishwasher **1**, and a third spray unit **43** arranged above the upper basket **53** in the height direction of the dishwasher **1**.
[0059] Each of the plurality of spray units **41**, **42**, and **43** may be configured to spray wash water while rotating. Each of the first spray unit **41**, the second spray unit **42**, and the third spray unit **43** may be provided to spray wash water while rotating. The plurality of spray units **41**, **42** and **43** may be referred to as a plurality of spray rotors. The first spray unit **41**, the second spray unit **42**, and the third spray unit **43**, respectively, may be referred to as a first spray rotor **41**, a second spray rotor **42**, and a third spray rotor **43**.
[0060] However, the spray device **40** may spray the wash water in a manner different from the above-described example. For example, unlike the second spray unit **42** and the third spray unit **43**, the first spray unit **41** may be fixed to one side of the lower surface **12***b* of the tub **12**. The first spray unit **41** may be configured to spray the wash water in a substantially horizontal direction by a fixed nozzle. A direction of the wash water, which is sprayed in a substantially horizontal direction from the nozzle of the first spray unit **41**, may be changed by a conversion assembly (not shown) arranged thereon and then the wash water may move upward. The conversion assembly may be installed on a rail by a holder and may be provided to be movable in translation along the rail.
[0061] The dishwasher **1** may include the sump assembly **70**.
[0062] The sump assembly **70** may be provided to receive wash water. The sump assembly **70** may collect wash water of the washing chamber C. For example, the lower surface **12***b* of the tub **12** may be inclined downward toward the sump assembly **70** to smoothly collect water to the sump assembly **70**. The wash water of the washing chamber C may flow along the slope of the lower surface **12***b* of the tub **12** and smoothly flow into the sump assembly **70**.
[0063] The sump assembly **70** may include a circulation pump **30** configured to pump wash water stored in the sump assembly **70** to the spray device **40**.
[0064] The sump assembly **70** may include a drain pump **60** configured to drain wash water and foreign substances (e.g., food residues) remaining in the sump assembly **70**.
[0065] The sump assembly **70** may pump the collected wash water and supply the wash water to the spray device **40**. The sump assembly **70** may be connected to the spray device **40** to supply wash water to the spray device **40**.
[0066] The sump assembly **70** may be independently connected to the first spray unit **41**, the second spray unit **42**, and the third spray unit **43**. For example, the sump assembly **70** may be independently connected to connectors connected to the first spray unit **41**, the second spray unit **42**, and the third spray unit **43**. The connector may be provided in the shape of a connection port, a duct, or the like.
[0067] For example, the second spray unit **42** and the third spray unit **43** may be provided with one connector, and in this case, wash water supplied through the one connector may flow into a connector. Wash water flowing into the connector may be branched during moving, and the branched wash water may be provided to at least one of the second spray unit **42** and the third spray unit **43**.
[0068] The dishwasher **1** may include an alternating device (not shown) configured to selectively supply wash water to the spray device **40**. The alternating device (not shown) may be driven to selectively supply wash water to each connector connected to each of the spray devices **41**, **42**, and **43**. For example, the alternating device (not shown) may selectively supply wash water to at least one of connectors connected to the first spray device **41** and a connector connected to the second spray device **42**.
[0069] The dishwasher **1** may include a machine room M, which is a space provided below the tub

12. The machine room M may be a place in which a component for circulating wash water is disposed.
[0070] For example, at least a part of the sump assembly **70** may be arranged in the machine room M. Most of the sump assembly **70** may be arranged in the machine room M.
[0071] The dishwasher **1** may include a detergent box **80** configured to input detergent into the tub **12**.
[0072] For example, the dishwasher **1** may include a display arranged on the front side of the dishwasher **1** to display a driving state of the dishwasher **1**. However, because the front panel **90** is disposed on the front of the dishwasher **1**, the display may be small or not exposed from the outside, and thus it is difficult for a user to recognize the display.
[0073] For example, due to the front panel **90** disposed on the front of the dishwasher **1**, the dishwasher **1** may not include a display configured to display a state to the front side of the dishwasher **1**.
[0074] For example, the dishwasher **1** may include a display apparatus **100** configured to display an image I on a floor B on which the dishwasher **1** is installed. As described above, the state of the dishwasher **1** may not be displayed forward due to the front panel **90** disposed on the front of the dishwasher **1**, and thus the display apparatus **100** may display the state of the dishwasher **1** by projecting the image I on the floor B.
[0075] For example, the display apparatus **100** may be configured to allow the image I to be display on the outside by projecting light, which is emitted from a light source, to an image.
[0076] For example, the display apparatus **100** may be disposed inside the door **20**.
[0077] For example, the display apparatus **100** may be disposed inside the outer surface **22** of the door **20**.
[0078] For example, the display apparatus **100** may be disposed on a rear side of the front panel **90**.
[0079] For example, the display apparatus **100** may be disposed outside the dishwasher **1**. The display apparatus **100** may receive information about the state of the dishwasher **1** through wireless communication and display the state of the dishwasher **1** as an image I.
[0080] For example, the display apparatus **100** may be disposed on the rear side of the front panel **90** and spaced apart from the dishwasher **1**.
[0081] For example, the display apparatus **100** may be disposed in a space different from the dishwasher **1**. The display apparatus **100** may receive information about the state of the dishwasher **1** through wireless communication and display the state of the dishwasher **1** as an image I. For example, when the dishwasher **1** is disposed in a kitchen of an internal space, the display apparatus **100** may be disposed in a living room. For example, the display apparatus **100** may be disposed on an art wall of a living room or on an inside of a wall forming the living room so as to project an image I on the floor of the living room.
[0082] However, hereinafter a type in which the display apparatus **100** is disposed inside the outer surface **22** will be described as an example.
[0083] FIG. **4** is a control block diagram of the dishwasher according to an embodiment, FIG. **5** is a conceptual diagram of a display apparatus according to an embodiment, FIG. **6** is a perspective view of the display apparatus according to an embodiment, FIG. **7** is a perspective view of the display apparatus according to an embodiment, and FIG. **8** is a cross-sectional view of the display apparatus according to an embodiment.
[0084] As illustrated in FIGS. **4** and **5**, the display apparatus **100** may include an image generator **1000** configured to generate an image I.
[0085] For example, the display apparatus **100** may include a display controller **900** configured to control the display apparatus **100**. The display controller **900** may transmit/receive a signal with a main controller **9** of the dishwasher **1** to control the display apparatus **100** to allow the display apparatus **100** to perform an operation.
[0086] For example, the display controller **900** may be provided as part of the main controller **9** of

the dishwasher **1** or provided as a separate component from the main controller **9** to communicate with the main controller **9** by wire or wirelessly.
[0087] The display controller **900** may include at least one memory and at least one processor to perform the above-described operation and an operation to be described later.
[0088] For example, the memory and the processor may be implemented as separate chips. For example, a processor may include one or more processor chips or one or more processing cores. For example, the memory may include one or two or more memory chips or one or two or more memory blocks. For example, the memory and the processor may be implemented as a single chip.
[0089] As illustrated in FIGS. **5** to **8**, the image generator **1000** may include a light source **1100**.
[0090] The image generator **1000** may include a display device **1200** configured to transmit light generated from the light source **1100** and configured to display an image i1.
[0091] The image generator **1000** may include an illumination lens **1300** configured to condense light generated from the light source **1100** toward the display device **1200**.
[0092] The image generator **1000** may include a reflector **1400** configured to guide light generated from the light source **1100** to the illumination lens **1300**.
[0093] The image generator **1000** may include a printed circuit board **1500** on which chips forming the above-described memory and processor are mounted.
[0094] The display apparatus **100** may include an adjustor **2000** including a refractive lens **2100** provided to adjust a size of an image formed by the image generator **1000**.
[0095] The adjustor **2000** may be configured to adjust the size and to focus of the image I projected on the floor B by adjusting the size of an image I displayed on the display device **1200**.
[0096] The adjustor **2000** may be configured to allow the image I, which has a larger size than the image I displayed on the display device **1200**, to be projected onto the floor B.
[0097] The adjustor **2000** may be configured to adjust the focus of the image I projected on the floor B to increase the user's visibility of the image I.
[0098] The adjustor **2000** may be configured to adjust the focus of the image I displayed on the floor B by moving relative to the image generator **1000**.
[0099] When an image generated by the image generator **1000** is transmitted through the adjustor **2000**, a distance between the image generator **1000** and the adjustor **2000** may be adjusted by the movement of the adjustor **2000**, and accordingly, the size and focus of the image I displayed on the floor B may be adjusted.
[0100] For example, the adjustor **2000** may be disposed below the image generator **1000** and may move relative to the image generator **1000** in the vertical direction. Accordingly, the size and focus of the image I displayed on the floor B may be adjusted.
[0101] The display apparatus **100** may include a housing **3000** in which the image generator **1000** and the adjustor **2000** are accommodated.
[0102] In the case of a display apparatus that projects an image, a projection distance of more than a predetermined distance from the display apparatus to display an image is required, and a predetermined light path lp**1** between the image generator and the adjustor is required to project an image according to the projection distance.
[0103] Accordingly, in the display apparatus, a housing is formed to cover the image generator, the adjustor, and the light path lp**1**, respectively. As the respective housings are spaced apart from each other, a volume of the display apparatus may be increased.
[0104] In the case of a display apparatus that is a projector type to display a state of a home appliance, the display apparatus is configured to project an image to a lower side of the home appliance and thus it is appropriate that the display apparatus is disposed in a lower portion of a front surface of the home appliance, particularly, an inside of a door of the home appliance or an inside of a panel of the home appliance forming the front surface.
[0105] At this time, when a volume of the display apparatus increases, it is difficult to arrange the display apparatus to the inside of the door or the inside of the panel of the home appliance.

[0106] When the light path lp**1** between the image generator and the adjustor is reduced or limited to a certain length for reducing the volume of the display apparatus, the resolution of the image displayed on the floor may be reduced because a focal range of the image displayed on the floor is limited. Further, a location on which the display apparatus is installed is limited, and thus there is difficulty in installing the display apparatus.
[0107] For example, the display apparatus **100** may be provided in such a way that the image generator **1000**, the adjustor **2000**, and the light path lp**1** between the image generator **1000** and the adjustor **2000** are all arranged in the housing **3000** formed as a unit. Therefore, it is possible to ease the above-mentioned difficulty.
[0108] The volume of the display apparatus **100** is minimized by the housing **3000** that is formed as a unit. Accordingly, the display apparatus **100** may be easily installed inside the door **20**.
[0109] The housing **3000** may be provided in such a way that the adjustor **2000** is configured to move relative to the image generator **1000** so as to change a distance between the image generator **1000** and the adjustor **2000**.
[0110] Accordingly, the resolution and size of the image I displayed on the floor B may be adjusted by adjusting the distance between the image generator **1000** and the adjustor **2000** even in various installation positions of the display apparatus **100** in the vertical direction. For example, the distance between the image generator **1000** and the adjustor **2000** may be adjusted in a plurality of steps or continuously.
[0111] Particularly, the housing **3000** may include a first housing **3100** on which the image generator **1000** is mounted and a second housing **3200** on which the adjustor **2000** is mounted.
[0112] For example, all components forming the image generator **1000** may be installed inside the first housing **3100**.
[0113] For example, all components forming the adjustor **2000** may be installed inside the second housing **3200**.
[0114] The second housing **3200** may be coupled to the first housing **3100** so as to move relative to the first housing **3100**.
[0115] For example, the second housing **3200** may be disposed below the first housing **3100**.
[0116] For example, the second housing **3200** may be coupled to the first housing **3100** from the lower side to the upper side of the first housing **3100** with respect to the vertical direction. For example, the first housing **3100** may be disposed above the second housing **3200** with respect to the vertical direction.
[0117] For example, the second housing **3200** may be provided to move with respect to the first housing **3100**.
[0118] For example, the second housing **3200** may move relative to the first housing **3100** with respect to the relative vertical direction.
[0119] Accordingly, all components forming the image generator **1000** are disposed inside the first housing **3100** and all components forming the adjustor **2000** are disposed inside the second housing **3200**. Further, the first housing **3100** and the second housing **3100** are configured to be coupled to each other. Therefore, the housing **3000** may accommodate all configurations of the display apparatus **100**, thereby minimizing the volume of the display apparatus **100**.
[0120] In addition, as the second housing **3200** moves relative to the first housing **3100**, the adjustor **2000** disposed inside the second housing **3200** may be configured to move relative to the image generator **1000** disposed inside the first housing **3100**. Therefore, it is possible to easily adjust the size and resolution of the image I displayed on the floor B.
[0121] Light may be provided to be emitted from the first housing **3100** to the second housing **3200** in an extension direction of an optical axis A (along the optical axis A), and the image I formed in the display apparatus **100** may be projected to the outside through the second housing **3200**. Accordingly, the second housing **3200** may include a projection hole **3220** through which the image I is projected from the display apparatus **100**.

[0122] For example, the projection hole **3220** may be provided to open downward. For example, the optical axis A may be formed to extend downward from the light source **1100** toward the projection hole **3220**.
[0123] The light source **1100**, the illumination lens **1300**, and the display device **1200** may be sequentially arranged from the top in the vertical direction inside the first housing **3100**.
[0124] The reflector **1400** may be disposed between the light source **1100** and the illumination lens **1300** with respect to the vertical direction.
[0125] For example, the light source **1100** may be provided to be mounted on the printed circuit board **1500**. For example, the light source **1100** may be provided to be mounted on a lower surface **1520** of the printed circuit board **1500** in the vertical direction. For example, in the first housing **3100**, the printed circuit board **1500** may be disposed above the reflector **1400**, the illumination lens **1300**, and the display device **1200** with respect to the vertical direction.
[0126] For example, the printed circuit board **1500**, on which the light source **1100** is mounted, the reflector **1400**, the illumination lens **1300**, and the display device **1200** may be sequentially disposed from the top in the vertical direction inside the first housing **3100**.
[0127] For example, the printed circuit board **1500** may be disposed inside the first housing **3100** to allow a surface, on which the light source **1100** is mounted, to face downward.
[0128] For example, because the second housing **3200** is disposed below the first housing **3100** with respect to the vertical direction, the refractive lens **2100** may be disposed below the light source **1100**, the illumination lens **1300**, and the display device **1200** with respect to the vertical direction.
[0129] The light source **1100**, the illumination lens **1300**, the display device **1200**, and the refractive lens **2100** may be sequentially disposed from the top in the vertical direction.
[0130] For example, because the light source **1100** is disposed on the lower surface **1520** of the printed circuit board **1500**, the light source **1100** may be provided to emit light downward. Light emitted from the light source **1100** may be provided to sequentially pass through the illumination lens **1300**, the display device **1200**, and the refractive lens **2100**.
[0131] Accordingly, the light source **1100**, the illumination lens **1300**, the display device **1200**, and the refractive lens **2100** may be sequentially arranged around the optical axis A of the light emitted from the light source **1100**.
[0132] The second housing **3200** may be configured to move relative to the first housing **3100** in the direction of the optical axis A. For example, the optical axis A of the display apparatus **100** may be formed from an upper side to a lower side according to the arrangement of the light source **1100**, and the second housing **3200** may be configured to move vertically.
[0133] In a state in which the light source **1100**, the illumination lens **1300**, the display device **1200**, and the refractive lens **2100** are sequentially arranged in the direction of the optical axis A, the refractive lens **2100** may be configured to move relative to the light source **1100**, the illumination lens **1300**, the display device **1200** along the optical axis A according to the relative movement of the second housing **3200**.
[0134] Because the second housing **3200** moves relative to the first housing **3100** with respect to the optical axis A, the adjustor **2000** may be configured to move relative to the image generator **1000** in a state in which the image generator **1000** disposed inside the first housing **3100** is fixed.
[0135] The image I may be projected from the display apparatus **100** to the floor B while light is transmitted through each component along the optical axis A. The size and resolution of the image I may vary according to a length of the light path lp**1** between the refractive lens **2100** and the display device **1200**, and a throw distance L that is from the display apparatus **100** to a point where the image I is projected. The point where the image I is projected from the display apparatus **100** is the floor B, and as the throw distance L is formed constant, the size and resolution of the image I may be adjusted according to the length of the light path lp**1** between the display device **1200** and the refractive lens **2100**.

[0136] As for the location of the display device **1200** and the reflective lens **2100** that determines the length of the light path lp**1**, the display apparatus **100** may be configured to allow the location of the reflective lens **2100** to move according to the movement of the second housing **3200** in a state in which the location of the display device **1200** is fixed.
[0137] Particularly, as the second housing **3200** moves relative to the first housing **3100** with respect to the optical axis A, the refractive lens **2100** may move relative to the display device **1200** with respect to the optical axis A. Accordingly, the light path lp**1** between the display device **1200** and the refractive lens **2100** may be easily changed through the movement of the second housing **3200**.
[0138] That is, the resolution and size of the image I displayed on the floor B may be adjusted by simply moving the second housing **3200** relative to the first housing **3100**, and accordingly, the volume of the display apparatus **100** may be minimized because an additional configuration or space for adjusting the resolution and size of the image I is unnecessary.
[0139] As illustrated in FIGS. **6** and **7**, the second housing **3200** may be provided in a barrel shape and disposed adjacent to or away from the first housing **3100** through rotation.
[0140] For example, the second housing **3200** may be rotated in one direction and moved upward, and accordingly, the second housing **3200** may be disposed adjacent to the first housing **3100** to reduce the light path lp**1**.
[0141] For example, the second housing **3200** may be rotated in the opposite direction and moved downward, and accordingly, the second housing **3200** may be disposed away from the first housing **3100** to increase the light path lp**1**.
[0142] For example, the first housing **3100** may include a female screw member **3110** including a screw thread formed at a predetermined angle, and the second housing **3200** may include a male screw member **3210** provided to be inserted into the female screw member **3110** and movable by including a screw thread corresponding to the screw thread of the female screw member **3110**.
[0143] A user can adjust the distance between the second housing **3200** and the first housing **3100** by rotating the second housing **3200**, and accordingly, the light path lp**1** may be adjusted and thus the focus of the image I may be adjusted.
[0144] While the second housing **3200** moves linearly on the extension direction of the optical axis A through the rotation of the second housing **3200**, the distance between the display device **1200** and the refractive lens **2100** may be changed and thus the focus of the image I may be adjusted.
[0145] In addition, the light source **1100** of the display apparatus **100** may be configured to emit light so as to supply light to the display device **1200** as the second housing **3200** moves in the vertical direction.
[0146] For example, the light source **1100** may be provided as a light emitting diode (LED).
[0147] A brightness of the image I may be determined according to an output of the light source **1100**, a light transmittance of the display device **1200**, the efficiency of light use in the adjustor **2000**, and the length of the projection distance between the display apparatus **100** and the floor B.
[0148] In addition, even when the image I maintains a constant brightness, the visibility of the image I may vary depending on the brightness around the floor B and the degree of reflectance or gloss of the floor B.
[0149] Accordingly, with the consideration that the brightness of the image I varies depending on the environment around the image I and that the length of the projection distance between the display apparatus **100** and the floor B varies, the light source **1100** may emit light with a predetermined output value to allow the image I to be displayed on the floor B with a brightness that is adjustable within a certain range.
[0150] For example, the light source **1100** may emit light to allow the brightness of the image I displayed on the floor B to be approximately 50 1× to 600 1×.
[0151] For example, the dishwasher **1** may include an illuminance sensor **85**. For example, the illuminance sensor **85** may be provided as one component of the display apparatus **100**.

[0152] For example, the illuminance sensor **85** may sense the brightness of the floor B on which the image I is displayed. For example, the display controller **9000** may control the output of the light source **1100** based on the value sensed by the illuminance sensor **85**.
[0153] The light source **1100** may be mounted on the printed circuit board **1500** to receive power, and the on/off and output levels of the light source **1100** may be controlled through the display controller **9000**. For example, the light source **1100** may be mounted on the lower surface **1520** of the printed circuit board **1500** with respect to the vertical direction and thus the light source **1100** may emit light downward.
[0154] For example, the printed circuit board **1500** may be arranged to allow mounting surfaces on both sides to be disposed in the vertical direction inside the first housing **3100**. For example, the printed circuit board **1500** may be disposed in such a way that a longitudinal direction of the printed circuit board **1500** is directed in the left and right direction Y.
[0155] For example, a chip may be disposed on an upper surface **1510** of the printed circuit board **1500** and the light source **1100** may be disposed on the lower surface **1520**.
[0156] For example, a cable **1530** provided to electrically connect a chip and the display device **1200** may be connected to the printed circuit board **1500**.
[0157] For example, the printed circuit board **1500** may be electrically connected to the main board forming the main controller **9** of the dishwasher **1** through the cable.
[0158] For example, the printed circuit board **1500** may be equipped with a communication device for wireless communication with the main board or may be connected to the communication device.
[0159] For example, through the cable, the printed circuit board **1500** may be electrically connected to a power supplier configured to supply power to the printed circuit board **1500**.
[0160] The reflector **1400** may be disposed between the light source **1100** and the illumination lens **1300** with respect to the vertical direction.
[0161] The reflector **1400** may be provided to reflect light, which is emitted from the surface of the light source **1100** at an angle greater than or equal to a predetermined angle, so as to guide the light to the illumination lens **1300**.
[0162] For example, the reflector **1400** may be provided in a cylindrical shape extending in the vertical direction and including a hollow.
[0163] For example, the reflector **1400** may be provided in a polygonal column shape extending in the vertical direction and including a hollow.
[0164] For example, the reflector **1400** may be provided in a shape in which a cross-sectional area increases from an upper opening **1410**, in which the light source **1100** is disposed, to a lower opening **1420** in which the illumination lens **1300** is disposed.
[0165] For example, the lower opening **1420** of the reflector **1400** may have a larger diameter than an aperture of the illumination lens **1300**. This is to prevent a case, in which a part of the light reflected inside the reflector **1400** is additionally reflected inside the reflector **1400** without being transmitted through the illumination lens **1300**, so as to prevent a reduction in the light efficiency.
[0166] For example, the inside of the reflector **1400** may be formed of a metal material.
[0167] For example, the inside of the reflector **1400** may be formed of a plastic material, and the outside of the reflector **1400** may be provided with a reflective coating.
[0168] For example, the inside of the reflector **1400** may be formed of a white material or painted in white.
[0169] The illumination lens **1300** may be provided to allow light to be uniformly emitted to an entire area of a display region of the display device **1200**.
[0170] For example, the illumination lens **1300** may be provided to allow light to be vertically transmitted through the display region of the display device **1200**.
[0171] For example, the illumination lens **1300** may be provided as a double-convex lens.
[0172] For example, the illumination lens **1300** may be provided to allow light to be emitted to the

display device **1200** with a uniformity of 90% or more over the entire display region of the display device **1200**. This is to minimize the luminance deviation of the image I projected on the floor B.
[0173] The light source **1100**, the illumination lens **1300**, and the display device **1200** may be sequentially arranged with respect to the optical axis A so as to allow the distance between the display device **1200** and the illumination lens **1300** to be less than the distance between the illumination lens **1300** and the light source **1100** with respect to the optical axis A. This is because it is easy for the light to vertically pass through the display region of the display device **1200** as the illumination lens **1300** is closer to the display device **1200**.
[0174] For example, when a lens surface, which faces the light source **1100**, of the illumination lens **1300** is defined as a first lens surface and a lens surface facing the display device **1200** is defined as a second lens surface, a curvature of the first lens surface may be provided to be greater than a curvature of the second lens surface. When the distance between the display device **1200** and the illumination lens **1300** is less than the distance between the illumination lens **1300** and the light source **1100**, it is easy for the light to vertically pass through the display region of the display device **1200**.
[0175] A light emitting area of the reflective lens **1300** with respect to a radial direction may be greater than an area of the display region of the display device **1200**. When a center of the illumination lens **1300** and a center of the display region of the display device **1200** are arranged on the same line based on the center of the optical axis A, the display region of the display device **1200** in the radial direction of the optical axis A may be arranged inside the illumination lens **1300**.
[0176] This is because the light is vertically emitted to the entire display region when the light emitting area of the reflective lens **1300** is greater than the area of the display region of the display device **1200**. It is appropriate that the light emitting area of the illumination lens **1300** is 1.1 to 1.5 times larger than the display region of the display device **1200**.
[0177] For example, the lens surface of the illumination lens **1300** may be provided as an aspheric surface. By adjusting a refractive index of light incident on an edge of the illumination lens **1300**, a light output efficiency of the edge of the illumination lens **1300** may be increased, and accordingly, the uniformity of the amount of light incident on the display region of the display device **1200** may be increased.
[0178] For example, the illumination lens **1300** may be formed of polycarbonate (PC) material.
[0179] For example, the illumination lens **1300** may be formed of Polymethyl methacrylate (PMMA) material.
[0180] For example, the illumination lens **1300** may be formed of a glass material.
[0181] The display device **1200** may be provided to display information of the image I. As the light emitted from the light source **1100** passes through the display device **1200**, an image displayed on the display device **1200** may be displayed on the image I projected onto the floor B.
[0182] For example, the display device **1200** may be provided as a liquid crystal display (LCD).
[0183] For example, the display device **1200** may be provided in the form of a film on which images, numbers, etc. are printed.
[0184] The display device **1200** may be provided to display a remaining operation time of the dishwasher **1**, a contamination level, a reservation status, an error, an operating state, and the like. The display device **1200** may be configured to receive numbers, images, symbols, etc. and to display the numbers, images, symbols, etc. on the image I.
[0185] For example, the display device **1200** may be provided as an LCD, and a color filter may be included in the LCD so as to display various colors on the image I.
[0186] For example, the display device **1200** may be provided with an LCD, and it is appropriate that a diameter or width of the LCD is provided approximately 10 mm, and the display apparatus **100** may be configured to allow the image I displayed on the floor to be projected with a diameter greater than the diameter of the LCD.
[0187] For example, when the diameter or width of the LCD is provided approximately 10 mm and

the light source **1100** is provided as an LED, the distance between the light source **1100** and the display region of the display device **1200** may be approximately 15 mm to 20 mm in consideration of the Lambertian distribution on a light emitting surface of the light source **1100**. This is to maximize light efficiency emitted from the light source **1100** in consideration of the Lambertian distribution. For example, the illumination lens **1300** may be disposed within a distance between the light source **1100** and the display region of the display device **1200**, and the distance between the illumination lens **1300** and the display region of the display device **1200** may be approximately 0.5 mm to 1.5 mm. This is to increase the efficiency of condensing light to the display device **1200** according to the refraction of the illumination lens **1300**.
[0188] For example, the display device **1200** may be provided as an LCD, and the display device **1200** may include a 7-segment LCD to display numbers. For example, in a 7-segment LCD, a total area of segments displaying numbers may be 75% or more of the total display region of the 7-segment LCD. This is to maintain the brightness of the image I above a predetermined value by securing the area of the segment in the 7-segment LCD to a predetermined value or more.
[0189] The refractive lens **2100** may be configured to adjust the size and resolution of the image I projected onto the floor B.
[0190] A curvature, a thickness, a material, a position of the refractive lens **2100** may be variously set according to the size of the display region of the display device **1200**, a target size of the image I, and a target resolution of the image I.
[0191] For example, the refractive lens **2100** may be provided in plurality according to the target size of the image I and the target resolution of the image I. For example, it is appropriate that the refractive lens **2100** includes 1 to 5 refractive lenses.
[0192] For example, the refractive lens **2100** may include a first refractive lens **2110** disposed most adjacent to the display apparatus, a second refractive lens **2120** disposed below the first refractive lens, and a third refractive lens **2130** disposed below the second refractive lens.
[0193] For example, the first refractive lens **2110** may be configured to allow light projected from the display device **1200** to travel in parallel while passing through the first refractive lens **2110**.
[0194] The first refracting lens **2110** may include a first lens surface **2111** facing the display device **1200** and a second lens surface **2112** facing the first lens surface **2111**.
[0195] For example, the first lens surface **2111** of the first refractive lens **2110** may be provided as a surface convex toward the display device **1200**. The second lens surface **2112** of the first refractive lens **2110** may be provided as a flat surface.
[0196] For example, the second refractive lens **2120** may be provided as a concave lens.
[0197] For example, the second refractive lens **2120** may be configured to allow a bundle of light, which passes through the first refractive lens **2110**, to expand.
[0198] For example, the third refractive lens **2130** may be provided as a convex lens.
[0199] For example, the third refractive lens **2130** may be configured to allow light, which passes through the second refractive lens **2120**, to be condensed and to allow the image I to be displayed on the floor B.
[0200] For example, the light passing through the display device **1200** may sequentially pass through the first refractive lens **2110**, the second refractive lens **2120**, and the third refractive lens **2130** and be projected onto the floor B.
[0201] For example, through the first refractive lens **2110**, the second refractive lens **2120**, and the third refractive lens **2130**, the display apparatus **100** may be configured to allow the image I to be display on the floor B with an area greater than the area of the display region of the display device **1200**.
[0202] For example, it is appropriate that each of the first refractive lens **2110**, the second refractive lens **2120**, and the third refractive lens **2130** has a thickness of 2.5 mm or more.
[0203] For example, the curvature of each lens surface of the first refractive lens **2110**, the second refractive lens **2120**, and the third refractive lens **2130** may be provided to be 10 mm or more.

[0204] For example, the refractive lens **2100** may be formed of Polycarbonate (PC) material.
[0205] For example, the refractive lens **2100** may be formed of Polymethyl methacrylate (PMMA) material.
[0206] For example, the refractive lens **2100** may be formed of a glass material.
[0207] For example, the refractive lens **2100** may include a Cyclic Olefin Copolymers (COC) material.
[0208] The dishwasher **1** may be not exposed to the outside of the front panel **90** for the aesthetic design, and thus a lower end **90***b* of the front panel **90** may be lower than a lower end **20***b* of the door **20** of the dishwasher **1** with respect to the vertical direction. Further, the display apparatus **100** may be disposed on the rear side of the front panel **90** in the front and rear direction, and as the display apparatus **100** is disposed inside the door **20**, the display apparatus **100** may be configured to emit light to the floor B from a position higher than the lower end **20***b* of the door **20** with respect to the vertical direction.
[0209] A portion of the light projected from the display apparatus **100** may be blocked by the lower end **20***b* of the door **20**, and thus a part of the image I projected on the floor B may not be displayed. To prevent this, when an area of the display region of the display device **1200** is defined as d**1**, a height y of the third refractive lens **2130** from the floor B, a height h of the third refractive lens **2130** from the lower end **90***b* of the front panel **90**, an aperture D of the third refractive lens **2130**, and a distance x between the aperture D of the third refractive lens **2130** and the center of the aperture D of the third refractive lens **2130** from the rear end **90***r* of the front panel **90** in the front and rear direction may be provided to satisfy an equation of $d1=2y/h*(x-D/2)+D$.
[0210] For example, D is defined as the aperture of the third refractive lens **2130**, but may be defined as an aperture of a lens including the largest aperture among the plurality of refractive lenses **2100**. For example, among the plurality of refractive lenses **2100**, the third refractive lens **2130** has the largest aperture, and thus D is defined as the aperture of the third refractive lens **2130**. However, when the second refractive lens **2120** has an aperture greater than the aperture of the third refractive lens **2130**, D may be defined as the aperture of the second refractive lens **2120**.
[0211] Hereinafter the housing **3000** will be described in detail.
[0212] FIG. **9** is an exploded perspective view of a housing of the display apparatus according to an embodiment, and FIG. **10** is an exploded perspective view of a second housing of the display apparatus according to an embodiment.
[0213] As described above, the display apparatus **100** may be provided in such a way that the image generator **1000**, the adjustor **2000**, and the light path lp**1** between the image generator **1000** and the adjustor **2000** are all arranged in the housing **3000** formed as a unit. Therefore, it is possible to minimize the volume of the display apparatus **100** and thus the display apparatus **100** may be disposed on the rear side of the front panel **90** or between the doors **20**.
[0214] The first housing **3100** may include a first case **3120** and a second case **3130** to allow components forming the image generator **1000** to be accommodated inside the first housing **3100**. The first case **3120** and the second case **3130** may be detachably coupled to each other. In a state in which the first case **3120** and the second case **3130** are separated, the components forming the image generator **1000** may be accommodated in the first case **3120** or the second case **3130**, and the components forming the image generator **1000** may be disposed in the first housing **3100** by coupling the first case **3120** and the second case **3130**.
[0215] For example, the first case **3120** and the second case **3130** may be coupled in the vertical direction. For example, the first case **3120** may be disposed on the upper side and the second case **3130** may be disposed on the lower side of the first case **3120**.
[0216] For example, the first case **3120** may be provided in such a way that an upper surface **3123** is sealed and a lower surface is open. For example, the second case **3130** may be coupled to the opened lower surface of the first case **3120**, thereby forming the first housing **3100**.
[0217] For example, because the upper surface **3123** of the first case **3120** is sealed, it is possible to

prevent moisture from entering the housing **3000** even when condensation occurs inside the door **20** of the dishwasher **1** on the upper side of the first housing **3100**.
[0218] For example, the printed circuit board **1500**, the light source **1100**, the illumination lens **1300**, the reflector **1400**, and the display device **1200** may be disposed inside the first case **3120**.
[0219] For example, the printed circuit board **1500** may be disposed in an upper portion of the inside of the first case **3120**. As the upper surface **3123** of the first case **3120** is sealed, it is possible to prevent external moisture from entering the printed circuit board **1500**.
[0220] For example, the first case **3120** may include a fixing member **3121** disposed inside the first case **3120**, and configured to fix the printed circuit board **1500**, the reflector **1400**, the illumination lens **1300**, and the display device **1200** inside the first case **3120**.
[0221] For example, the printed circuit board **1500** may be fixed to an upper side of the fixing member **3121**.
[0222] For example, the light source **1100**, the illumination lens **1300**, and the reflector **1400** may be disposed inside the fixing member **3121**.
[0223] For example, the display device **1200** may be fixed to the lower side of the fixing member **3121**.
[0224] For example, the fixing member **3121** may be provided to be coupled with the second case **3130**.
[0225] Each component may be fixed to the inside of the first case **3120** by the fixing member **3121**, and as the second case **3130** and the first case **3120** are coupled, each component may be disposed inside the first housing **3100** while being fixed to the fixing member **3121**.
[0226] For example, the first case **3120** may include an outlet **3125** for discharging heat generated from the light source **1100**. The printed circuit board **1500** and the light source **1100** may be damaged because the temperature inside the first housing **3100** rises due to the heat generated from the light source **1100**. However, it is possible to circulate the heat through the outlet **3125** and to reduce the internal temperature of the first housing **3100** by heat dissipation through the heat circulation.
[0227] For example, the outlet **3125** may be disposed at a higher position than the board **1520** to enable heat circulation. For example, the outlet **3125** may be provided in a shape in which a part of the first case **3120** is opened on an upper part of the first case **3120** so as to be disposed adjacent to the light source **1100**. This is because the printed circuit board **1500** is disposed on the inner upper side of the first case **3120**.
[0228] For example, an eave **3126** may be provided on the upper side of the outlet **3125** to prevent penetration of moisture from the outside.
[0229] For example, a chamfer **3124** may be included between the upper surface **3123** of the first case **3120** and a side surface extending downward from an edge of the upper surface **3123**. This is to prevent moisture from accumulating on the upper surface **3123** when moisture flows into the display apparatus **100** due to water leakage inside the door **20**. The chamfer **3124** may be provided to allow moisture introduced into the upper surface **3123** to flow to the lower portion of the first case **3120** through the chamfer **3124**.
[0230] The female screw member **3110** including a screw thread formed at a predetermined angle may be disposed in the second case **3130**.
[0231] For example, the female screw member **3110** may be disposed in the lower portion of the second case **3130** and open downward.
[0232] The female screw member **3110** may include a hollow **3111** to allow the male screw member **3210** to be inserted into the female screw member **3110** and rotated. Light transmitted from the display device **1200** may be emitted to the second housing **3200** through the hollow **3111**.
[0233] For example, the female screw member **3110** may be provided to extend from the upper end of the second case **3130** to the rear end. For example, the female screw member **3110** may be provided in a cylindrical or polygonal shape including the hollow **3111**. For example, the display

device **1200** may be disposed on the upper end of the female screw member **3110** to allow light transmitted from the display device **1200** to be directly emitted into the hollow **3111** of the female screw member **3110**.

[0234] For example, the second housing **3200** may be provided in a barrel shape. For example, the second housing **3200** may be provided in a cylindrical or polygonal shape including a hollow.

[0235] The male screw member **3210** may be disposed in the upper portion of the second housing **3200** and thus when the male screw member **3210** is inserted into the female screw member **3110**, the second housing **3200** may be coupled to the first housing **3100**.

[0236] The second housing **3200** may include a lens accommodating member **3270** disposed under the male screw member **3210** and provided to accommodate the plurality of refractive lenses **2110**, **2120**, and **2130** inside the second housing **3200**.

[0237] For example, the second housing **3200** may include a plurality of grooves **3250** extending in the direction of the optical axis A on an outer circumferential surface in which the lens accommodating member **3270** is disposed.

[0238] The plurality of grooves **3250** may have a shape **3250** in which a concave-convex shape or a groove-protrusion shape are repeated on the outer circumferential surface of the second housing **3200** in a circumferential direction of the second housing **3200** and provided to be disposed spaced apart from each other to be longer than an adjustable length lp**2** along the direction of the optical axis A.

[0239] A user can adjust the height of the second housing **3200** by rotating the second housing **3200** to focus the image I. When the second housing **3200** is rotated, the male screw member **3210** may move upward or downward through the screw thread in the female screw member **3110**, and thus the length of the light path lp**1** may be adjusted.

[0240] The second housing **3200** may include a handle **3260** that protrudes in a radial direction of the second housing **3200** to allow a user to easily grip the second housing **3200**.

[0241] For example, the handle **3260** may have an annular shape and more protrude than the outer circumferential surface of the lens accommodating member **3270** in the radial direction of the second housing **3200**.

[0242] For example, the handle **3260** may protrude in the radial direction of the second housing **3200**.

[0243] For example, the handle **3260** may be disposed lower than a stopper **3140** to be described later inserted into the lower portion of the plurality of concavo-convex grooves **3250** in the direction of the optical axis A.

[0244] For example, the handle **3260** may be disposed below the lens accommodating member **3270** in the vertical direction. For example, the handle **3260** may be disposed above the lens accommodating member **3270** in the vertical direction. For example, the handle **3260** may be disposed on the lens accommodating member **3270** in the vertical direction.

[0245] The male screw member **3210** may move upward or downward on the female screw member **3110** through rotation by the adjustable length lp**2**, and thus the length of the light path lp**1** may be adjusted by the change in the length of the adjustable length lp**2**.

[0246] The adjustable length lp**2** may be substantially equal to or less than the length of the male screw member **3210** in the vertical direction.

[0247] The concave-convex grooves **3250** may be provided equal to or greater than the focus adjustable length lp**2** in the direction of the optical axis A in the vertical direction.

[0248] The first housing **3100** may include the stopper **3140** provided to restrict the arbitrary rotation of the second housing **3200** relative to the first housing **3100**.

[0249] For example, the stopper **3140** may be detachably provided in the first housing **3100**. For example, the stopper **3140** may be provided to be coupled to the second case **3130**. For example, the stopper **3140** may be integrally formed with the second case **3130**. Alternatively, the stopper **3140** may be integrally formed with the first case **3120**.

[0250] For example, one or more stoppers **3140** may be provided. For example, the stopper **3140** may be provided as a pair. For example, the pair of stoppers **3140** may be in contact with both ends of the outer circumferential surface of the lens accommodating member **3270** in the direction of 180 degrees and thus the pair of stoppers **3140** may prevent the second housing **3200** from being rotated beyond the user's intended position.
[0251] For example, three or more stoppers **3140** may be provided. For example, the plurality of stoppers **3140** may be spaced apart from each other in a circumferential direction of an outer circumferential surface of the lens accommodating member **3270** and provided to be in contact with the lens accommodating member **3270**.
[0252] For example, the stopper **3140** may prevent the second housing **3200** from being arbitrarily rotated by a force applied to the second housing **3200**.
[0253] For example, a user can rotate the second housing **3200** by applying a greater force to the lens accommodating member **3270** than a force due to tension between the stopper **3140** and the lens accommodating member **3270**.
[0254] For example, the stopper **3140** may include an arm **3141** extending in the direction of the optical axis A and an insertion protrusion **3142** disposed on the lower side of the arm **3141** and provided to be inserted into one of plurality of concave-convex grooves **3250**.
[0255] For example, the arm **3141** may be formed of an elastic material. A state in which the insertion protrusion **3142** is inserted into the groove may be maintained by elasticity of the arm **3141** in a state in which the insertion protrusion **3142** is inserted into one of the plurality of grooves **3250**. At this time, unless a user rotates the lens accommodating member **3270** with a force greater than the tension at which the insertion protrusion **3142** is inserted into the groove according to the elasticity of the arm **3141**, the insertion protrusion **3142** may be maintained in a state of being inserted into the groove.
[0256] For example, the insertion protrusion **3142** may be inserted into one of the plurality of grooves **3250**, thereby restricting the arbitrary rotation of the second housing **3200**.
[0257] When the second housing **3200** is rotated by the user's pressure, the insertion protrusion **3142** may be separated from one of the plurality of grooves **3250**, and when the rotation of the second housing **3200** is finished, the insertion protrusion **3142** may be inserted into any other groove, which is the closest to the insertion protrusion **3142** among the plurality of grooves **3250** after the rotation of the second housing **3200**, so as to restrict the arbitrary rotation of the second housing **3200**.
[0258] For example, when the insertion protrusion **3142** moves away from a high part of the outer circumferential surface of the lens accommodating member **3270** through rotation of the second housing **3200**, the insertion protrusion **3142** may collide with a groove due to the elasticity of the arm **3141** of the stopper **3140** and a noise caused by the collision may be generated. Accordingly, a user can adjust the amount of rotation of the second housing **3200** by recognizing the sound during the adjusting process.
[0259] The second housing **3200** may include a first case **3230** and a second case **3240** to accommodate the plurality of refractive lenses **2110**, **2120**, and **2130** in the second housing **3200**.
[0260] The first case **3230** and the second case **3240** may be provided to extend in the direction of the optical axis A and coupled to each other in a direction perpendicular to the optical axis A. This is to easily accommodate the plurality of refractive lenses **2110**, **2120**, and **2130** in the first case **3230** or the second case **3240**.
[0261] When the first case **3230** and the second case **3240** are provided to be coupled in the direction of the optical axis A, it is required that the plurality of refractive lenses **2110**, **2120**, and **2130** may be accommodated in a stacked manner inside the first case **3230** or the second case **3240**. However, in a process of stacking the plurality of refractive lenses **2110**, **2120**, and **2130**, a tolerance in the position of the plurality of refractive lenses **2110**, **2120**, and **2130** may occur, thereby causing the displacement of the optical axis A.

[0262] However, the first case **3230** and the second case **3240** may be provided to extend in the direction of the optical axis A and coupled to each other in the direction perpendicular to the optical axis A. Accordingly, in a state in which the first case **3230** and the second case **3240** are separated, it is possible to accommodate the plurality of refractive lenses **2110**, **2120**, and **2130** in the direction perpendicular to the direction of the optical axis A in the first case **3230** or the second case **3240**, and then the first case **3230** and the second case **3240** may be coupled to each other. Therefore, it is possible to allow the plurality of refractive lenses **2110**, **2120**, and **2130** to be accommodated in designated positions inside the second housing **3200**.
[0263] The first refractive lens **2110** may include a first refractive lens holder **2111** surrounding an edge of the first refractive lens **2110**. The lens accommodating member **3270** may include a first refractive lens insertion groove **3271**, to which the first refractive lens holder **2111** is inserted, so as to allow the first refractive lens to be accommodated in the second housing **3200**.
[0264] The second refractive lens **2120** may include a second refractive lens holder **2121** surrounding an edge of the second refractive lens **2120**. The lens accommodating member **3270** may include a second refractive lens insertion groove **3272**, to which the second refractive lens holder **2121** is inserted, so as to allow the second refractive lens to be accommodated in the second housing **3200**.
[0265] The third refractive lens **2130** may include a third refractive lens holder **2131** surrounding an edge of the third refractive lens **2130**. The lens accommodating member **3270** may include a third refractive lens insertion groove **3273**, to which the third refractive lens holder **2131** is inserted, so as to allow the third refractive lens to be accommodated in the second housing **3200**.
[0266] In the manufacturing process, because the size of the plurality of refractive lenses **2110**, **2120**, and **2130** is small, each of the refractive lenses **2110**, **2120**, and **2130** may be inserted into different refractive lens insertion grooves without being inserted into the corresponding refractive lens insertion grooves **3271**, **3272**, and **3273**, and thus the plurality of refractive lenses **2110**, **2120**, and **2130** may be disposed in the wrong order in the vertical direction.
[0267] To prevent the above-mentioned difficulty, each of the refractive lens holders **2111**, **2121**, and **2131** may include a mark having a different size or shape.
[0268] The first refractive lens holder **2111** may include a first mark **2112**.
[0269] The second refractive lens holder **2121** may include a second mark **2122**.
[0270] The third refractive lens holder **2131** may include a third mark **2132**.
[0271] The first refractive lens insertion groove **3271** may have a shape corresponding to the first mark **2112** to allow the first mark **2112** to be inserted therein.
[0272] The second refractive lens insertion groove **3272** may have a shape corresponding to the second mark **2122** to allow the second mark **2122** to be inserted therein.
[0273] The third refractive lens insertion groove **3273** may have a shape corresponding to the third mark **2132** to allow the third mark **2132** to be inserted therein.
[0274] In other words, the marks **2112**, **2122**, and **2132** are formed in different shapes or sizes, and thus unless the marks **2112**, **2122**, and **2132** are inserted to the corresponding refractive lens insertion grooves **3271**, **3272**, and **3273**, the marks **2112**, **2122**, and **2132** may not be inserted into the refractive lens insertion grooves **3271**, **3272**, and **3273**. Accordingly, each of the refractive lenses **2110**, **2120**, and **2130** may be easily accommodated in a predetermined position. In a home appliance including a display apparatus configured to project an image displaying state information of the home appliance according to an embodiment, the display apparatus may include a light source and a display device configured to transmit light generated by the light source and display an image.
[0275] For example, the display apparatus may include an image generator including an illumination lens configured to condense the light generated by the light source to the display device.
[0276] For example, the display apparatus may include an adjustor including a refractive lens

configured to adjust a size of the image projected from the display apparatus.
[0277] For example, the display apparatus may include a housing in which the image generator and the adjustor are accommodated.
[0278] For example, the light source, the illumination lens, the display device, and the refractive lens may be sequentially arranged in an extension direction of an optical axis formed by the image generator.
[0279] For example, the refractive lens may be configured to move in the extension direction of the optical axis relative to the light source, the illumination lens, and the display device.
[0280] The configurations inside the display apparatus may be sequentially arranged in the extension direction of the optical axis, so that the volume of the housing may be reduced, and accordingly, the display apparatus may be easily disposed in a narrow space.
[0281] For example, the optical axis may be configured to extend in a vertical direction.
[0282] For example, the refractive lens may be configured to move in the vertical direction relative to the light source, the illumination lens, and the display device.
[0283] For example, the display apparatus may be configured to project the image larger than an image displayed on the display device onto a floor on which the home appliance is installed.
[0284] For example, the housing may include a first housing in which the image generator is accommodated and a second housing in which the adjustor is accommodated.
[0285] For example, the second housing may be configured to move relative to the first housing.
[0286] For example, the display apparatus may be configured such that the image transmitted through the refractive lens is projected downward from the housing.
[0287] For example, the second housing may be configured to move in the extension direction of the optical axis relative to the first housing to allow a distance between the display device and the refractive lens to be changeable.
[0288] For example, the second housing may be configured to be reciprocally movable in the extension direction of the optical axis with respect to the first housing by rotation.
[0289] For example, the refractive lens may be provided as a plurality of refractive lenses.
[0290] For example, the plurality of refractive lenses may be configured to be sequentially arranged in the extension direction of the optical axis inside the second housing.
[0291] For example, the plurality of refractive lenses may include a first lens disposed closest to the image generator in the extension direction of the optical axis and on which the light transmitted from the display device is incident, the first lens comprising a first surface having a convex shape and a second surface having a flat shape and facing the first surface.
[0292] For example, the plurality of refractive lenses may include a second lens disposed next to the first lens in the extension direction of the optical axis and provided as a concave lens.
[0293] For example, the plurality of refractive lenses may include a third lens disposed next to the second lens in the extension direction of the optical axis and provided as a convex lens.
[0294] For example, when an area of the image displayed onto a floor on which the home appliance is installed is defined as d**1**, an aperture of the refractive lens disposed at a lower end among the plurality of refractive lenses is defined as D, a height from the aperture of the refractive lens to a lower end of an outer panel is defined as h, a height of the aperture of the refractive lens disposed at the lower end from the floor is defined as y, and a distance from a center of the aperture of the refractive lens disposed at the lower end to a rear end of the outer panel in a front and rear direction is defined as x, the home appliance may be configured to satisfy an equation of $d1=2y/h*(x-D/2)+D$.
[0295] For example, the display apparatus may be arranged to a rear side of a front panel arranged on a front surface of the home appliance.
[0296] For example, the display apparatus may be configured to project the image downward.
[0297] For example, the home appliance may further include a door having a front surface formed by the front panel and configured to open an inner space of the home appliance forward to a front.

[0298] For example, the display apparatus may be disposed inside the door.
[0299] For example, in a dishwasher including a washing chamber, a door configured to open and close forwardly of the washing chamber, and a display apparatus configured to project an image displaying state information of the dishwasher onto a floor on which the dishwasher is installed, the display apparatus may be disposed inside the door.
[0300] For example, the display apparatus may include an image generator including a light source, a display device configured to transmit light generated by the light source and display an image, and an illumination lens configured to condense the light generated by the light source to the display device.
[0301] For example, the display apparatus may include an adjustor including a refractive lens configured to adjust a size of the image projected from the display apparatus.
[0302] For example, the display apparatus may include a housing in which the image generator and the adjustor are accommodated.
[0303] For example, the light source, the illumination lens, the display device, and the refractive lens may be sequentially arranged in a vertical direction.
[0304] For example, the refractive lens may be configured to move in the vertical direction relative to the light source, the illumination lens, and the display device.
[0305] For example, the housing may include a first housing in which the image generator is accommodated.
[0306] For example, the housing may include a second housing in which the adjustor is accommodated.
[0307] For example, the second housing is configured to move in an extension direction of the optical axis relative to the first housing to allow a distance between the display device and the refractive lens to be changeable.
[0308] For example, the refractive lens may be provided as a plurality of refractive lenses.
[0309] For example, the plurality of refractive lenses may be sequentially arranged in the vertical direction inside the second housing.
[0310] For example, the door may include a front surface forming an appearance of the door, and a rear surface facing an inner space when the door closes the inner space.
[0311] For example, the display apparatus may be disposed between the front surface of the door and the rear surface of the door in a front and rear direction.
[0312] For example, in a display apparatus configured to project an image, the display apparatus may include an image generator including a light source, a display device configured to transmit light generated by the light source and display an image, and an illumination lens configured to condense the light generated by the light source to the display device.
[0313] For example, the display apparatus may include an adjustor including a plurality of refractive lenses configured to adjust a size of the image displayed on the display device to allow a size of the image projected from the display apparatus to be adjustable.
[0314] For example, the display apparatus may include a housing in which the image generator and the adjustor are accommodated.
[0315] For example, the light source, the illumination lens, the display device, and the refractive lens may be sequentially arranged in an extension direction of an optical axis formed by the image generator.
[0316] For example, the refractive lens may be configured to move in the extension direction of the optical axis relative to the light source, the illumination lens, and the display device.
[0317] For example, the housing may include a first housing in which the image generator is accommodated.
[0318] For example, the housing may include a second housing in which the adjustor is accommodated.
[0319] For example, the second housing may be configured to move in the extension direction of

the optical axis relative to the first housing.
[0320] For example, the refractive lens may be provided as a plurality of refractive lenses.
[0321] For example, the plurality of refractive lenses may be sequentially arranged in the extension direction of the optical axis inside the second housing.
[0322] Although a few embodiments of the disclosure have been shown and described, it would be appreciated by those skilled in the art that changes may be made in these embodiments without departing from the principles and spirit of the disclosure, the scope of which is defined in the claims and their equivalents.
[0323] While the present disclosure has been particularly described with reference to exemplary embodiments, it should be understood by those of skilled in the art that various changes in form and details may be made without departing from the spirit and scope of the present disclosure.

Claims

1. A home appliance comprising: a display apparatus configured to project an image including state information of the home appliance, wherein the display apparatus comprises: an image generator comprising a light source, a display device configured to transmit light generated by the light source and display the image, and an illumination lens configured to condense the light generated by the light source to the display device, an adjustor comprising a refractive lens configured to adjust a size of the image projected from the display apparatus, and a housing configured to accommodate the image generator and the adjustor, wherein the light source, the illumination lens, the display device, and the refractive lens are sequentially arranged along an optical axis of the image generator, and the refractive lens is configured to move along the optical axis relative to the light source, the illumination lens, and the display device.
2. The home appliance of claim 1, wherein the optical axis is along a vertical direction of the home appliance, and the refractive lens is configured to move along the vertical direction relative to the light source, the illumination lens, and the display device.
3. The home appliance of claim 1, wherein the display apparatus is configured to project the image larger than an image displayed on the display device onto a floor on which the home appliance is installed.
4. The home appliance of claim 1, wherein the housing comprises a first housing in which the image generator is accommodated, and a second housing in which the adjustor is accommodated, and the second housing is configured to move relative to the first housing and to allow the image transmitted through the refractive lens to be projected downward from the housing.
5. The home appliance of claim 4, wherein the second housing is configured to move along the optical axis relative to the first housing to allow a distance between the display device and the refractive lens to be changeable.
6. The home appliance of claim 4, wherein the second housing is configured to be reciprocally movable along the optical axis with respect to the first housing by rotation.
7. The home appliance of claim 4, wherein the refractive lens is among a plurality of refractive lenses, and the plurality of refractive lenses are configured to be sequentially arranged along the optical axis inside the second housing.
8. The home appliance of claim 7, wherein the plurality of refractive lenses comprises: a first lens disposed closest to the image generator along the optical axis and on which the light transmitted from the display device is incident, the first lens comprising a first surface having a convex shape and a second surface having a flat shape and facing the first surface, a second lens disposed next to the first lens along the optical axis and provided as a concave lens, and a third lens disposed next to the second lens along the optical axis and provided as a convex lens.
9. The home appliance of claim 7, wherein based on an area of the image displayed onto a floor on which the home appliance is installed is defined as d**1**, an aperture of the refractive lens disposed at

a lower end among the plurality of refractive lenses is defined as D, a height from the aperture of the refractive lens to a lower end of an outer panel is defined as h, a height of the aperture of the refractive lens disposed at the lower end from the floor is defined as y, and a distance from a center of the aperture of the refractive lens disposed at the lower end to a rear end of the outer panel in a front and rear direction is defined as x, an equation of $d1=2y/h*(x-D/2)+D$ is satisfied.

10. The home appliance of claim 1, wherein the display apparatus is arranged to a rear side of a front panel arranged on a front surface of the home appliance, and configured to project the image downward.

11. The home appliance of claim 1, further comprising a door having a front surface formed by a front panel and configured to open forwardly and to allow access to an inner space of the home appliance, wherein the display apparatus is disposed inside the door.

12. A dishwasher comprising: a washing chamber; a door configured to be moveable forward of the washing chamber to open and close the washing chamber; and a display apparatus configured to project an image including state information of the dishwasher onto a floor on which the dishwasher is installed, wherein the display apparatus is configured to be disposed inside the door; wherein the display apparatus comprises: an image generator comprising a light source, a display device configured to transmit light generated by the light source and display the image, and an illumination lens configured to condense the light generated by the light source to the display device, an adjustor comprising a refractive lens configured to adjust a size of the image projected from the display apparatus, and a housing configured to accommodate the image generator and the adjustor, wherein the light source, the illumination lens, the display device, and the refractive lens are sequentially arranged along a vertical direction of the dishwasher.

13. The dishwasher of claim 12, wherein the refractive lens is configured to move along the vertical direction relative to the light source, the illumination lens, and the display device.

14. The dishwasher of claim 12, wherein the housing comprises a first housing configured to accommodate the image generator and a second housing configured to accommodate the adjustor.

15. The dishwasher of claim 14, wherein the second housing is configured to move along an optical axis of the image generator relative to the first housing to allow a distance between the display device and the refractive lens to be changeable.